

CPSC 4970 – C++ Programming

Homework 7

Your solution to each problem should include the C++ source code and the screenshot of a test run. A solution without a test run will have points deducted.

1. (The Date Class) Design a class called `Date`. The class should store a date in three integers: `month`, `day`, and `year`. There should be member functions to print the date in the following forms:

12/25/2022

December 25, 2022

25 December 2022

The class should also implement the following exception classes:

`InvalidDay` Throw when an invalid day (< 1 or > 31) is passed to the class.

`InvalidMonth` Throw when an invalid month (< 1 or > 12) is passed to the class.

Demonstrate the class in a driver program.

2. (Minimum/Maximum Templates) Write templates for the two functions `minimum` and `maximum`. The `minimum` function should accept two arguments and return the value of the argument that is the lesser of the two. The `maximum` function should accept two arguments and return the value of the argument that is the greater of the two. Design a simple driver program that demonstrates the templates with various data types.

3. (Word Frequency) Write a program that reads the contents of a text file. The program should create a map in which the keys are the individual words found in the file and the values are the number of times each word appears. For example, if the word “the” appears 128 times, the map would contain an element with “the” as the key and 128 as the value. The program should display the frequency of each word.

4. (Word Index) Write a program that reads the contents of a text file. The program should create a map in which the key–value pairs are described as follows:

- **Key**—The keys are the individual words found in the file.
- **Values**—Each value is a set that contains the line numbers in the file where the word (the key) is found.

For example, suppose the word “robot” is found in lines 7, 18, 94, and 138. The map would contain an element in which the key was the string “robot”, and the value was a set containing the numbers 7, 18, 94, and 138. Once the map is built, the program should create another text file, known as a word index, listing the contents of the map. The word index file should contain an alphabetical listing of the words that are stored as keys in the map, along with the line numbers where the words appear in the original file. The figure below shows an example of an original text file and its index file.

Kennedy.txt - Notepad

File Edit Format View Help

We observe today not a victory
of party but a celebration
of freedom symbolizing an end
as well as a beginning
signifying renewal as well
as change

Ln 12, Col

index.txt - Notepad

File Edit Format View Help

We: 1
a: 1 2 4
an: 3
as: 4 5 6
beginning: 4
but: 2
celebration: 2
change: 6
end: 3
freedom: 3
not: 1
observe: 1
of: 2 3
party: 2
renewal: 5
signifying: 5
symbolizing: 3
today: 1
victory: 1
well: 4 5

Ln 21, Col 1